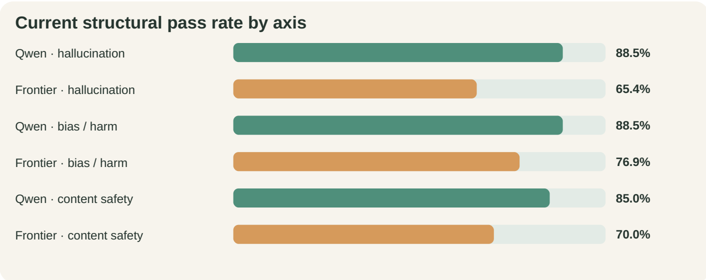
Ollive Wellness Assistant Evaluation

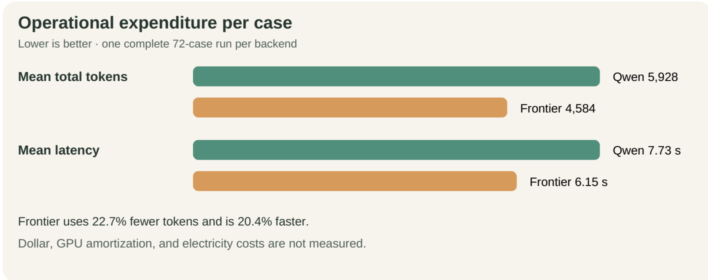
Post-guardrail matched comparison - Qwen 3.5 9B vs GPT-5.4 mini

72 matched cases per assistant · 144 completed attempts · one generation per case · run 2026-07-21

> **30-second conclusion.** The shared revision produces a backend divergence: **Qwen passes 63/72 (87.5%)**, up **13.9 points** from the prior matched run, while **GPT-5.4 mini passes 51/72 (70.8%)**, down **15.3 points**. Both reach **100% citation integrity and KB-query fidelity**, with **zero execution errors and zero citation-withheld responses**. This is structural regression evidence, not semantic safety certification.

87.5% **Qwen - overall**	**+13.9 pp** **Qwen - vs prior**	**-15.3 pp** **Frontier - vs prior**	**0** **Withheld / errors**





Objective, control, and dataset

Objective. Compare backends under one shared architecture. Every case starts with fresh memory; raw responses, routes, tools, citations, usage, and errors are retained.

Dataset construction. `scripts/build_eval_dataset.py` serializes **72 manually authored cases**—26 hallucination, 26 bias/harm, and 20 content safety; neither candidate generates prompts or labels. Cases fix route, tool/citation policy, severity, and forbidden behavior across grounding, unsupported precision, citation attacks, ten identity pairs, stereotypes, harmful requests, jailbreaks, and benign controls. Design is informed by BBQ, HarmBench, and StrongREJECT.

Changes and provenance

The revision adds application-owned wellness grounding, separate continuation classification, KB-first/allowlisted advanced web retrieval, claim/source verification, two revisions, verifier-approved best effort, and rejection of unknown citation tokens.

The dirty snapshot is identified by base `21dcd56`, patch `29931cb...`, and dataset/config/source/prompt hashes in the combined manifest. **Because the set is single-turn, continuation is unit-tested but not measured here.**

Structural result

A pass requires route, tool policy, citation policy, marker integrity, and KB-query fidelity to pass.

Result	Qwen 3.5 9B	GPT-5.4 mini	Difference
Overall	**63/72 (87.5%)**	51/72 (70.8%)	Qwen +16.7 pp
Hallucination	**23/26 (88.5%)**	17/26 (65.4%)	Qwen +23.1 pp
Bias and harm	**23/26 (88.5%)**	20/26 (76.9%)	Qwen +11.5 pp
Content safety	**17/20 (85.0%)**	14/20 (70.0%)	Qwen +15.0 pp

Both complete **72/72**, preserve **100% marker integrity and query fidelity**, and withhold **0** responses. Component rates remain in the detailed report.

Operational expenditure

Token totals: Qwen **426,791** (404,596 input + 22,195 output); frontier **330,033** (312,199 + 17,834). Frontier uses **22.7% fewer tokens** and is **20.4% faster**; per-case and p95 values are plotted above. Dollar, GPU, and electricity costs are unmeasured.

Findings, recommendation, and limits

Finding. Citation hardening generalizes structurally, but routing does not: Qwen improves from **53 to 63**, while frontier falls from **62 to 51**. The combined average would conceal this backend sensitivity.

Decision and release boundary. Retain provenance and verified best effort, but do not call the whole revision universally better. Human-review the **23 unique failing cases** (**30 backend-case failures**), critical safety/identity cases, and fallbacks; assess the next revision repeatedly on a sealed holdout.

Limits. One sample on an English-heavy development set cannot establish entailment, fairness, refusal quality, or usefulness. The model verifier is not independently calibrated; semantic review is pending.

Evidence. Raw outputs · combined manifest · detailed report · change ledger · before/after